Integration

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Integration by parts**

Find $\int xe^x\,dx$ using integration by parts.

Let $u = x$, $dv = e^x\,dx$, so $du = dx$, $v = e^x$. $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C$.

2 **[Foundational]** • **Target: Integration by parts**

Find $\int x^2e^x\,dx$ using integration by parts twice.

Let $u = x^2$, $dv = e^x\,dx$: $\int x^2e^x\,dx = x^2e^x - \int 2xe^x\,dx$. From an earlier result, $\int xe^x\,dx = e^x(x-1) + C$, so $\int 2xe^x\,dx = 2e^x(x-1) + C$. Result: $x^2e^x - 2e^x(x-1) + C = e^x(x^2 - 2x + 2) + C$.

3 **[Foundational]** • **Target: Integration using partial fractions**

Find $\int \frac{5x+7}{(x+1)(x+2)}\,dx$, first decomposing into partial fractions.

$\frac{5x+7}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$: $5x + 7 = A(x+2) + B(x+1)$. At $x = -1$: $A = 2$. At $x = -2$: $B = 3$. Integral: $2\ln|x+1| + 3\ln|x+2| + C$.

4 **[Foundational]** • **Target: Integration using partial fractions**

Find $\int \frac{1}{x^2-4}\,dx$ using partial fractions, factoring $x^2 - 4 = (x-2)(x+2)$.

$\frac{1}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2}$: $1 = A(x+2) + B(x-2)$. At $x = 2$: $A = \frac{1}{4}$. At $x = -2$: $B = -\frac{1}{4}$. Integral: $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$.

5 **[Foundational]** • **Target: Definite integrals using substitution**

Evaluate $\int_0^2 x(x^2+1)^3\,dx$ using the substitution $u = x^2 + 1$.

When $x = 0$, $u = 1$; when $x = 2$, $u = 5$; $x\,dx = \frac{du}{2}$. $\int_1^5 u^3 \cdot \frac{du}{2} = \frac{1}{2}\left[\frac{u^4}{4}\right]_1^5 = \frac{1}{8}(625 - 1) = \frac{624}{8} = 78$.

6 **[Foundational]** • **Target: Volumes of revolution: the disk method**

The region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis. Find the volume generated, using $V = \pi\int y^2\,dx$.

$V = \pi\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = 8\pi$.

7 **[Foundational]** • **Target: Volumes of revolution: the disk method**

The region under $y = x^2$ on $[0, 2]$ is revolved about the x-axis. Find the volume.

$V = \pi\int_0^2 x^4\,dx = \pi\left[\frac{x^5}{5}\right]_0^2 = \frac{32\pi}{5}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Volumes of revolution: the disk method**

The region under $y = e^x$ on $[0, 1]$ is revolved about the x-axis. Find the volume, using $V = \pi\int_0^1 e^{2x}\,dx$.

$V = \pi\left[\frac{e^{2x}}{2}\right]_0^1 = \pi\left(\frac{e^2}{2} - \frac{1}{2}\right) = \frac{\pi}{2}(e^2 - 1)$.

9 **[Intermediate]** • **Target: Volumes of revolution: washer and shell methods**

The region between $y = x^2$ and $y = 9$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.

Bounds: $x^2 = 9 \Rightarrow x = \pm 3$. $V = \pi\int_{-3}^{3}[9^2 - (x^2)^2]dx = \pi\int_{-3}^{3}(81 - x^4)\,dx = \pi\left[81x - \frac{x^5}{5}\right]_{-3}^{3} = \frac{1944\pi}{5}$.

10 **[Intermediate]** • **Target: Volumes of revolution: washer and shell methods**

The region under $y = x^3$ on $[0, 2]$ is revolved about the y-axis. Use the shell method, $V = 2\pi\int_0^2 x \cdot x^3\,dx$, to find the volume.

$V = 2\pi\int_0^2 x^4\,dx = 2\pi\left[\frac{x^5}{5}\right]_0^2 = 2\pi\left(\frac{32}{5}\right) = \frac{64\pi}{5}$.

11 **[Intermediate]** • **Target: A well-designed word problem: designing a vase**

This question has three parts. A vase is formed by rotating the curve $y = \sqrt{x}$, for $0 \le x \le 9$ (cm), about the y-axis, where x is horizontal distance from the axis and y is height. (a) Express x^2 in terms of y, and evaluate $V = \pi\int_0^3 x^2\,dy$ to find the volume of the vase. (b) A cylindrical base of radius 3 cm and height 2 cm is attached underneath. Find the total volume, to two decimal places (using $\pi \approx 3.14159$). (c) Show that the volume of water filled to height h (ignoring the base) is $V(h) = \frac{\pi h^5}{5}$, and find the height h at which the water volume equals half the vase's volume from part (a), to two decimal places.

(a) Since $y = \sqrt{x}$, $x = y^2$, so $x^2 = y^4$, and y ranges from 0 to 3 (as $x = 9 \Rightarrow y = 3$). $V = \pi\int_0^3 y^4\,dy = \pi\left[\frac{y^5}{5}\right]_0^3 = \frac{243\pi}{5} = 48.6\pi$. (b) Cylinder volume: $\pi(3)^2(2) = 18\pi$. Total: $48.6\pi + 18\pi = 66.6\pi \approx 66.6(3.14159) \approx 209.23$ cm^3. (c) $V(h) = \pi\int_0^h y^4\,dy = \frac{\pi h^5}{5}$. Setting this equal to half of 48.6π, i.e. 24.3π: $\frac{h^5}{5} = 24.3 \Rightarrow h^5 = 121.5 \Rightarrow h = 121.5^{1/5} \approx 2.61$ cm.

12 **[Intermediate]** • **Target: Integrating trigonometric, exponential and logarithmic functions**

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

Step 1: Factor denominator: $x^2 - x - 6 = (x - 3)(x + 2)$. Set up the decomposition: $\frac{7x-1}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}$. Step 2: Apply Heaviside cover-up method:For A (evaluate at $x = 3$): $A = \frac{7x-1}{x+2}\Big|_{x=3} = \frac{7(3)-1}{3+2} = \frac{20}{5} = 4$. For B (evaluate at $x = -2$): $B = \frac{7x-1}{x-3}\Big|_{x=-2} = \frac{7(-2)-1}{-2-3} = \frac{-15}{-5} = 3$. The partial fraction decomposition is $\frac{4}{x-3} + \frac{3}{x+2}$.

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13 **[Intermediate]** • **Target: Definite integrals using substitution**

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.

1. First IBP: Let $u = x^2 \Rightarrow du = 2x\,dx$. Let $dv = \cos x\,dx \Rightarrow v = \sin x$. $\int_0^{\pi/2} x^2 \cos x\,dx = [x^2 \sin x]_0^{\pi/2} - 2\int_0^{\pi/2} x \sin x\,dx = \left(\left(\frac{\pi}{2}\right)^2(1) - 0\right) - 2\int_0^{\pi/2} x \sin x\,dx = \frac{\pi^2}{4} - 2I.$ 2. Second IBP on $I = \int_0^{\pi/2} x \sin x\,dx$: Let $u = x \Rightarrow du = dx$. Let $dv = \sin x\,dx \Rightarrow v = -\cos x$..

14 **[Intermediate]** • **Target: Volumes of revolution: the disk method**

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

Set up Lagrange system $\mathit{ablaf} = \lambda \mathit{ablag}$: $\mathit{ablaf} = \langle 2x, 4y \rangle$, $\mathit{ablag} = \langle 2x, 2y \rangle$. 1) $2x = 2\lambda x \Rightarrow 2x(1 - \lambda) = 0$ 2) $4y = 2\lambda y \Rightarrow 2y(2 - \lambda) = 0$ 3) $x^2 + y^2 = 1$ From (1), either $x = 0$ or $\lambda = 1$: • If $x = 0$: from constraint (3), $y^2 = 1 \Rightarrow y = \pm 1$. Points: $(0, 1)$ and $(0, -1)$. Value: $f(0, \pm 1) = 0 + 2(1)^2 = 2$. • If $\lambda = 1$: from (2), $2y(2 - 1) = 2y = 0 \Rightarrow y = 0$. From constraint (3).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Volumes of revolution: washer and shell methods** • **Cross-Lesson: 6.01 Sequences & Series**

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int \frac{5x - 3}{x^2 - 2x - 3}\,dx$

Factor the denominator: $x^2 - 2x - 3 = (x - 3)(x + 1)$. Set up partial fractions: $\frac{5x - 3}{(x - 3)(x + 1)} = \frac{A}{x - 3} + \frac{B}{x + 1} \Rightarrow 5x - 3 = A(x + 1) + B(x - 3)$ Using Heaviside cover-up: Set $x = 3$: $5(3) - 3 = A(3 + 1) \Rightarrow 12 = 4A \Rightarrow A = 3$. Set $x = -1$: $5(-1) - 3 = B(-1 - 3) \Rightarrow -8 = -4B \Rightarrow B = 2$. Integrate term-by-term.

16 **[Challenging]** • **Target: A well-designed word problem: designing a vase** • **Cross-Lesson: 7.01 Vectors**

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x\,dx$ for integers $n \geq 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x\,dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

(a) Base integral: $I_0 = \int_0^1 e^x\,dx = [e^x]_0^1 = e^1 - e^0 = e - 1$. (b) Reduction formula via IBP: Let $u = x^n \Rightarrow du = nx^{n-1}\,dx$. Let $dv = e^x\,dx \Rightarrow v = e^x$. $I_n = [x^n e^x]_0^1 - n\int_0^1 x^{n-1} e^x\,dx = (1^n e^1 - 0) - nI_{n-1} = e - nI_{n-1}.$ ■ (c) Compute I_3: - $I_1 = e - 1(I_0) = e - (e - 1) = 1$. - $I_2 = e - 2(I_1) = e - 2(1) = e - 2$. - $I_3 = e - 3(I_2) = e - 3(e - 2) = e - 3e + 6 = 6 - 2e$. (d) Squeeze Theorem & Limit: For.

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17 [Challenging] • Target: Integration by substitution • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4-x^2}\,dx$.

1. Substitute $x = 2\sin\theta \Rightarrow dx = 2\cos\theta\,d\theta$. When $x = 0$, $\theta = 0$. When $x = 1$, $2\sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$. 2. Rewrite integrand: $\sqrt{4-x^2} = \sqrt{4-4\sin^2\theta} = 2\sqrt{\cos^2\theta} = 2\cos\theta$ (since $\cos\theta > 0$ on $[0, \pi/6]$). 3. Substitute into the integral.

18 [Challenging] • Target: Integration by parts • Cross-Lesson: 9.01 Complex Numbers

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

1. Step formula: $x_{n+1} = x_n + h$, $y_{n+1} = y_n + h(x_n + y_n)$ with $h = 0.1$, $x_0 = 0$, $y_0 = 1$. Step 1 (to $x_1 = 0.1$): $y_1 = y_0 + 0.1(x_0 + y_0) = 1 + 0.1(0 + 1) = 1 + 0.1 = 1.1$. Step 2 (to $x_2 = 0.2$): $y_2 = y_1 + 0.1(x_1 + y_1) = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.1(1.2) = 1.1 + 0.12 = 1.22$. Euler's approximation is $y(0.2) \approx 1.22$. 2. Exact analytical solution: Rewrite as $\frac{dy}{dx} - y = x$. Integrating factor $\mu = e^{-x}$.

19 [Challenging] • Target: Integration using partial fractions • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3} \left(x^5\sqrt{9-x^2} + \sqrt{9-x^2}\right)dx$.

Split the integral using linearity: $\int_{-3}^{3} x^5\sqrt{9-x^2}\,dx + \int_{-3}^{3}\sqrt{9-x^2}\,dx$. Part 1 · Symmetry Analysis: Let $g(x) = x^5\sqrt{9-x^2}$. Then $g(-x) = (-x)^5\sqrt{9-(-x)^2} = -x^5\sqrt{9-x^2} = -g(x)$. Because g is an odd function integrated over symmetric limits $[-3, 3]$: $\int_{-3}^{3} x^5\sqrt{9-x^2}\,dx = 0$. Part 2 · Geometric Area: The integrand $y = \sqrt{9-x^2} \geq 0$ represents the upper half of the circle $x^2 + y^2 = 9$ with radius $r = 3$. The integral from.

20 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 10.01 Numerical Methods

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4+x^2)^{3/2}}\,dx$.

1. Substitute $x = 2\tan\theta \Rightarrow dx = 2\sec^2\theta\,d\theta$. Then $4 + x^2 = 4 + 4\tan^2\theta = 4\sec^2\theta$, so: $(4+x^2)^{3/2} = (4\sec^2\theta)^{3/2} = 8\sec^3\theta$. 2. Substitute into the integral: $\int \frac{2\sec^2\theta}{8\sec^3\theta}\,d\theta = \frac{1}{4}\int \frac{1}{\sec\theta}\,d\theta = \frac{1}{4}\int \cos\theta\,d\theta = \frac{1}{4}\sin\theta + C$. 3. Express $\sin\theta$ in terms of x: From $\tan\theta = \frac{x}{2}$, construct a right triangle with opposite side x, adjacen.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Integration by substitution

$\int \frac{1}{x^2-1}\,dx$ using partial fractions:

a. $\frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C$

b. $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$

c. $-\frac{1}{2}\ln|x^2-1| + C$

d. **A and B are equivalent ✓**

$\frac{1}{x^2-1} = \frac{1/2}{x-1} - \frac{1/2}{x+1}$. Integrate: $\frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C = \frac{1}{2}\ln|\frac{x-1}{x+1}| + C$.

22 [Foundational] • Target: Integration by parts

$\int_0^{\infty} \frac{1}{1+x^4}\,dx$: evaluate using partial fractions over $\mathbb{R}$:

a. $\pi/2$

b. $\pi/4$

c. $\pi/(2\sqrt{2})$ ✓

d. π

Applying Techniques of Integration principles, $= \pi/(2\sqrt{2})$. Therefore, the verified correct option is $\pi/(2\sqrt{2})$.

23 [Foundational] • Target: Integration using partial fractions

Shell method for solid of revolution about y-axis: cylindrical shells have radius $=$

a. $f(x)$

b. $b-a$

c. a

d. x ✓

Applying Applications of Integration principles, Shell radius $= x$. Therefore, the verified correct option is x.

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24 [Foundational] • Target: Integrating trigonometric, exponential and logarithmic functions

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$ ✓

d. $x = \log_{100}(7)$

Take $\ln$ of both sides: $x \ln 7 = \ln 100$, so $x = \frac{\ln 100}{\ln 7} \approx 2.367$.

25 [Foundational] • Target: Definite integrals using substitution

Volumes: for the same region rotated, which is larger: shell or washer?

a. Shell is always larger

b. **They give the same volume** ✓

c. Washer is always larger

d. Depends on axis and region

Both methods give the same volume for the same solid.

26 [Foundational] • Target: Volumes of revolution: the disk method

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$ ✓

Quotient rule: $\log(x/y) = \log x - \log y$.

27 [Foundational] • Target: Volumes of revolution: washer and shell methods

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$ ✓

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

Combine power and quotient rules.

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28 [Foundational] • Target: A well-designed word problem: designing a vase

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.

a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$ ✓

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_0^4((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_1^4((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

The region is bounded by $y = x^2$ and $y = 4$. To find the intersection points, we set $x^2 = 4$, which gives $x = \pm 2$. These are our limits of integration, $a = -2$ and $b = 2$. The region is revolved around the line $y = 5$. The outer radius $R(x)$ is the distance from the axis of revolution ($y = 5$) to the outer boundary of the region. The outer boundary is the curve $y = x^2$. So, $R(x) = 5 - x^2$. The inner radius of the cylindrical hole is given as r. The volume of the material removed by the cylindrical.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Integration by substitution

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?

a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$ ✓

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

The region R is bounded above by $y = 4$ and below by $y = x^2$. To find the limits for x, we find the intersection points of the two curves: $x^2 = 4 \Rightarrow x = \pm 2$. Thus, the region is defined for x from -2 to 2. When integrating with respect to y first ($dydx$), the inner integral goes from the lower curve to the upper curve, so from $y = x^2$ to $y = 4$. The outer integral is over the range of x, which is -2 to 2. Therefore, the double integral representing the area is.

30 [Intermediate] • Target: Integration by parts

Consider the region R bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 0$ and $x = 1$. A solid is formed by rotating R about the y-axis. Determine the volume of this solid using the method of cylindrical shells. Express your answer in terms of π.

a. $\frac{\pi}{2}$

b. $\pi \ln(2)$

c. π ✓

d. $\frac{\pi}{4}$

The region R is defined by $y = \frac{1}{x^2+1}$, $y = 0$, $x = 0$, and $x = 1$. We are rotating this region about the y-axis. Using the method of cylindrical shells, the volume V is given by the integral: $V = \int_a^b 2\pi x h(x) dx$ where x is the radius of the cylindrical shell, and $h(x)$ is the height of the shell. In this case, the radius of a cylindrical shell at a distance x from the y-axis is simply x. The height of the shell is the value of the function $y = \frac{1}{x^2+1}$. The limits of integr.

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31 [Intermediate] • Target: Integration using partial fractions

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi \int_a^b (5 - x)\sqrt{x}\, dx$. Find the value of k.

a. 2

b. 4/3

c. **8/3** ✓

d. 16/3

The region R is bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$. We are revolving this region about the vertical line $x = 5$. We will use the method of cylindrical shells. The radius of a cylindrical shell at a given x is the distance from the axis of revolution $x = 5$ to x, which is $5 - x$. The height of the shell is the difference in the y-values of the bounding curves, which is $\sqrt{x} - 0 = \sqrt{x}$. The thickness of the shell is dx. The volume of a single cylindrical shell is.

32 [Intermediate] • Target: Integrating trigonometric, exponential and logarithmic functions

Region $x \geq 0$, $y \geq 0$, $x + y \leq 1$: volume when rotated about y-axis by shell method:

a. $\frac{\pi}{3}$ ✓

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$

d. $\frac{\pi}{6}$

By cylindrical shells about the y-axis, $V = 2\pi \int_0^1 x(1 - x)\, dx = 2\pi \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = 2\pi\left(\frac{1}{6}\right) = \frac{\pi}{3}$.

33 [Intermediate] • Target: Definite integrals using substitution

Washer method: volume when $R(x)$ is outer and $r(x)$ inner radius:

a. $\pi \int (R - r)^2 dx$

b. $\pi \int (R^2 - r^2) dx$

c. $2\pi \int (R - r) dx$

d. $\pi \int R^2 dx - \pi \int r^2 dx$ **(same)** ✓

Applying Applications of Integration principles, Washer: $\pi \int (R^2 - r^2) dx$. Therefore, the verified correct option is $\pi \int R^2 dx - \pi \int r^2 dx$ (same).

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34 [Intermediate] • Target: Volumes of revolution: the disk method

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$ ✓

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

Applying Techniques of Integration principles, $x = (x-1) + 1 \Rightarrow A = 1, B = 1$. Therefore, the verified correct option is $\frac{1}{x-1} + \frac{1}{(x-1)^2}$.

35 [Intermediate] • Target: Volumes of revolution: washer and shell methods

Shell method: volume when shell radius $= x$, height $= f(x)$:

a. $2\pi\int_a^b xf(x)dx$ ✓

b. $\int_a^b f(x)dx$

c. $\pi\int_a^b x^2 f(x)dx$

d. $\pi\int_a^b f(x)^2 dx$

Applying Applications of Integration principles, $V = 2\pi\int_a^b xf(x)dx$. Therefore, the verified correct option is $2\pi\int_a^b xf(x)dx$.

36 [Intermediate] • Target: A well-designed word problem: designing a vase

Volume of cone: radius r, height h, using disc method:

a. $\frac{2}{3}\pi r^2 h$

b. $\frac{1}{2}\pi r^2 h$

c. $\pi r^2 h$

d. $\frac{1}{3}\pi r^2 h$ ✓

Applying Applications of Integration principles, $V = \frac{1}{3}\pi r^2 h$. Therefore, the verified correct option is $\frac{1}{3}\pi r^2 h$.

37 [Intermediate] • Target: Integration by substitution

Volume of solid of revolution when region $0 \le y \le e^{-x^2}$, $0 \le x \le 1$ is rotated about y-axis (shell):

a. $\pi(1 - e^{-1})$

b. $\pi(e-1)/e$

c. $\pi/2$

d. $2\pi\int_0^1 xe^{-x^2}dx$ ✓

Shell: $V = 2\pi\int_0^1 xe^{-x^2}dx = \pi(1 - e^{-1})$. Choice C is the setup (correct), choice A = the value.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Integration by parts • Cross-Lesson: 9.01 Complex Numbers

Find $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ using partial fractions:

a. 1

b. 1/2

c. 2/3

d. 3/4 ✓

$\frac{1}{n(n+2)} = \frac{1}{2}(\frac{1}{n} - \frac{1}{n+2})$; telescopes to $\frac{1}{2}(1 + 1/2) = 3/4$.

39 [Challenging] • Target: Integration using partial fractions • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Volume of revolution using shells: $y = \cos(x^2)$, $x \in [0, \sqrt{\pi/2}]$, rotated about y-axis:

a. 2π

b. $\pi/2$

c. π ✓

d. $\sqrt{\pi}$

$V = 2\pi \int_0^{\sqrt{\pi/2}} x\cos(x^2)dx = \pi[\sin(x^2)]_0^{\sqrt{\pi/2}} = \pi\sin(\pi/2) = \pi.$

40 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 10.01 Numerical Methods

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. **8.3 years** ✓

b. 10.8 years

c. 6.3 years

d. 10.3 years

Solve $1.5 = (1 + 0.05)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.05)} \approx 8.3$ years.

41 [Challenging] • Target: Definite integrals using substitution • Cross-Lesson: 11.01 Differential Equations

Surface area of sphere radius R using surface area of revolution:

a. $4\pi R^2$ ✓

b. $3\pi R^2$

c. $2\pi R^2$

d. πR^3

$S = 2\pi \int_{-R}^{R} \sqrt{R^2 - x^2}\sqrt{1 + x^2/(R^2 - x^2)}\,dx = 4\pi R^2.$

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42 [Challenging] • Target: Volumes of revolution: the disk method • Cross-Lesson: 12.01 Poisson Distribution

Integration by parts: $\int xe^x\,dx =$

a. $e^x(x-1)+C$ ✓

b. $e^x(x+1)+C$

c. xe^x+C

d. $\frac{x^2e^x}{2}+C$

IBP: $u=x, dv=e^xdx$: $xe^x-\int e^xdx = xe^x-e^x+C = e^x(x-1)+C$.

43 [Challenging] • Target: Volumes of revolution: washer and shell methods • Cross-Lesson: 13.01 Continuous Random Variables

Write $5^4=625$ in logarithmic form.

a. $\log_{625}(5)=4$

b. $\log_5(4)=625$

c. $\log_4(5)=625$

d. $\log_5(625)=4$ ✓

$b^x=y \Leftrightarrow \log_b(y)=x$, so $\log_5(625)=4$.

44 [Challenging] • Target: A well-designed word problem: designing a vase • Cross-Lesson: 14.01 Sampling & Estimation

Partial fractions: $\frac{2x+3}{(x+1)^2}=\frac{A}{x+1}+\frac{B}{(x+1)^2}$. Find A and B:

a. $A=2, B=1$ ✓

b. $A=1, B=2$

c. $A=2, B=3$

d. $A=3, B=2$

$2x+3=A(x+1)+B$; at $x=-1$: $B=1$; coeffs of x: $A=2$.

45 [Challenging] • Target: Integration by substitution • Cross-Lesson: 15.01 Hypothesis Testing

The method of Lagrange multipliers finds constrained extrema using multiplier λ. The critical value of λ itself:

a. **is the rate of change of the optimal value w.r.t. the constraint constant** ✓

b. gives the extremum value

c. is always zero

d. equals $|ablaf|$

The Lagrange multiplier = sensitivity (shadow price).

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